

**This assignment will not be graded and is only for practice.**

**Level 1**

1) Find the rank of the following matrix:  $\begin{bmatrix} 0 & 1 & 3 & 1 \\ 1 & 0 & 0 & 1 \\ 3 & 2 & 0 & 3 \end{bmatrix}$

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

**1 point**

Consider the following set of vectors  $S$  in  $\mathbb{R}^3$  to answer the questions 2, 3 and 4.  
 $S = \{(1, 2, 0), (0, 3, 1), (3, 3, -1), (3, 0, -2)\}$ .

2) Let  $V$  be the vector space spanned by the vectors  $(1, 2, 0)$  and  $(0, 3, 1)$ . Which one **1 point** of the following is correct?

- ☐  $V = \{(a, 2a + 3b, b) \mid a, b \in \mathbb{R}\}$
- ☐  $V = \{(0, 2a - 3b, b) \mid a, b \in \mathbb{R}\}$
- ☐  $V = \{(-a, 2a + 3b, 0) \mid a, b \in \mathbb{R}\}$
- ☐  $V = \{(0, 2a + 3b, 0) \mid a, b \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \{(a, 2a + 3b, b) \mid a, b \in \mathbb{R}\}$

3) If the vectors in  $S$  are written as the columns of a matrix  $A$ , then what will be the rank of  $A$ ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

4) What will the dimension of the vector space spanned by  $S$  be?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

5) Choose the set of correct options from the following.

**1 point**

- ☐ Row rank and column rank of a matrix is always the same.
- ☐ The rank of a zero matrix is always 0.
- ☐ The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.
- ☐ Rank of the  $n \times n$  identity matrix is 1 for any  $n \in \mathbb{N}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Row rank and column rank of a matrix is always the same.

The rank of a zero matrix is always 0.

The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.

**Level 2**

6) Consider the following upper triangular matrix to choose the correct options.

**1 point**

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

where  $a, b, c, d, e, f \in \mathbb{R}$ .

- ☐ If  $f = 0$ , then the rank of the matrix must be less than or equal to 2.
- ☐ If  $f = 0$ , then the rank of the matrix must be exactly 2.
- ☐ If  $a, b, c, d, e, f$  are all non-zero then the rank of the matrix must be 3.
- ☐ If  $a, d, f$  are non-zero then the rank of the matrix must be 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If  $f = 0$ , then the rank of the matrix must be less than or equal to 2.

If  $a, b, c, d, e, f$  are all non-zero then the rank of the matrix must be 3.

If  $a, d, f$  are non-zero then the rank of the matrix must be 3.

7) If rank of the matrix  $\begin{bmatrix} 2 & -3 & 4 \\ 0 & 1 & -2 \\ 1 & -3 & a \end{bmatrix}$  is 2 then find the value of  $a$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

**1 point**

8) Find the rank of the matrix  $A$ , where  $A = [a_{ij}]$  is of order  $3 \times 3$  and  $a_{ij} = \min\{i, j\}$ ,  $i, j = 1, 2, 3$ .

[Hint: Write down the matrix and apply row operations to derive it's reduced row echelon form.]

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

**1 point**

9) What is the rank of the matrix  $A = \begin{bmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$ , where  $x, y, z$  are distinct non-zero integers?

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

**1 point**

10) Rank of a  $3 \times 3$  matrix  $A$  is 3 and another  $3 \times 3$  matrix  $B$  is 2. What is the minimum possible value for the rank of  $AB$ ?

---

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

Your score is: 0/10